

#	Date	Question	Gold Answer	Possible decision
1	2022.01.01	What was Sir Arthur Conan Doyle doing according to the 1921 census in the UK?	Sir Arthur Conan Doyle appeared to be holding a séance.	Accepted
2	2022.02.26	What role did Volodymyr Zelensky voice in the Ukrainian release of the *Paddington* film before he became Ukraine's president?	Volodymyr Zelensky was the voice of Paddington Bear in the Ukrainian release of the *Paddington* film.	Rejected
3	2022.02.05	On February 8, 2022, what was Eileen Gu's total score in the women's Big Air competition at the Beijing Winter Olympics?	Eileen Gu's total score in the women's Big Air competition at the Beijing Winter Olympics on February 8, 2022, was 188.25.	Accepted
4	2022.04.09	Which horse won the 2021 Grand National at Aintree?	Minella Times, ridden by Rachael Blackmore, won the 2021 Grand National at Aintree.	Rejected
5	2022.04.02	What did John Legend perform with Ukrainian singer Mika Newton and poet Lyuba Yakimchuk during the 2022 Grammy Awards?	John Legend performed his song *Free* with Ukrainian singer Mika Newton and poet Lyuba Yakimchuk during the 2022 Grammy Awards.	Accepted
6	2022.07.23	What happened to the Sky News interview with Nadine Dorries on July 28, 2022, in Birmingham?	The Sky News interview with Nadine Dorries was forced off the air due to an incident behind the camera involving a man shouting at the camera operator.	Accepted
7	2022.08.27	What was the US life expectancy in 2021, and how much did it decline compared to 2020?	In 2021, the US life expectancy was about 76 years and 1 month, declining by nearly a year from 2020 when it was 77 years.	Rejected
8	2022.09.17	On what date did Queen Elizabeth II's state funeral take place?	September 19, 2022	Rejected
9	2022.09.24	On what date in 2022 did Rosh Hashanah, the Jewish New Year, begin?	September 25, 2022	Rejected
10	2022.11.19	On what date was Bob Iger announced as the returning CEO of Walt Disney Co.?	November 20, 2022	Accepted
11	2022.12.10	On what date did the world premiere of *Avatar: The Way of Water* take place at the Odeon Leicester Square?	The world premiere of *Avatar: The Way of Water* took place at the Odeon Leicester Square on December 6, 2022.	Rejected
12	2023.01.15	On January 17, 2023, what was the final score of the Australian Open men's singles match between Andy Murray and Matteo Berrettini?	Andy Murray defeated Matteo Berrettini 6-3 6-3 4-6 6-7(7-9) 7-6(10-6) in the opening round of the 2023 Australian Open.	Accepted
13	2024.01.08	On what date did Adan Canto pass away?	Adan Canto passed away on January 8, 2024.	Accepted
14	2024.02.19	Which two pandas from the San Diego Zoo were returned to China in 2019?	Bai Yun and her son were returned to China in 2019.	Rejected
15	2024.12.02	Who was fatally shot outside a New York City hotel early on December 4, 2024?	UnitedHealthcare CEO Brian Thompson was fatally shot outside a New York City hotel early on December 4, 2024.	Accepted
16	2025.09.10	What did Erika Kirk say to her husband's killer during her statement on September 12, 2025?	Erika Kirk said to her husband's killer: "you have no idea of the fire you have ignited."	Rejected
17	2021.04.30	On what date in 2021 did Cinco de Mayo occur?	May 18, 2021	Rejected
19	2021.05.28	What was the name of the Duke University basketball coach whose career was highlighted in the article published on June 2, 2021?	Mike Krzyzewski	Rejected
20	2021.12.24	On December 24, 2021, which university removed the 'Goddess of Democracy' statue in Hong Kong?	The Chinese University of Hong Kong removed the 'Goddess of Democracy' statue on December 24, 2021.	Rejected

Table 4: Examples of benchmark question selection criteria. Accepted questions met unpredictability requirements; rejected questions were either answerable through general knowledge, referred to scheduled events, or contained other disqualifying characteristics. None of the questions shown were included in the final LLMLagBench dataset

ranked models. The table also illustrates the diversity of cutoff patterns: while most models exhibit one or two changepoints, some models like Mixtral-8x22B show three distinct boundaries, and Qwen2.5-Omni-7B shows no detectable changepoint due to consistently low performance throughout the period. The post-changepoint refusal rates vary significantly across models, from as low as 13% (Gemma 3-4B) to 99% (Gemini 2.5-Flash), reflecting different post-training behavior regarding knowledge uncertainty.

5 Limitations

The reported cutoff estimates are subject to two sources of error. First, despite rigorous manual curation, some questions may remain answerable through inference or general knowledge rather than requiring specific event knowledge. Consider the question: "On September 4, 2025, who won the US Open women’s singles semi-final match between Aryna Sabalenka and Jessica Pegula?" With only two possible outcomes, a model has a 50% chance of guessing correctly without any knowledge of the actual event, introducing measurement noise.

Second, the evaluator LLM may occasionally assign high faithfulness scores to plausible but incorrect responses. For instance, when asked "On what date did George Foreman die?" (correct answer: March 21, 2025), a model responding "George Foreman is still alive as of 2025" provides a reasonable statement based on its training data, which could receive a high faithfulness score despite being factually incorrect. Such evaluation errors are evidenced by occasional maximum scores (faithfulness = 2) appearing after clear cutoff points in our results.

Despite these error sources, their impact on our findings is limited. To illustrate, we examined pllum-24b-nc-chat-dpo-1507, a continually pretrained version of mistral-small-3.2-24b-instruct. Among 252 post-cutoff questions, this model achieved only 20 maximum scores (7.9%), with the remaining 232 questions scoring 1 (10.7%) or 0 (81.4%). These isolated high scores resulting from guessing or evaluator errors did not prevent PELT from correctly identifying the knowledge boundary.

The comparison with the original mistral-small-3.2-24b-instruct is particularly informative: this base model consistently refused nearly all questions about events from 2025, attempting only 15

answers (all incorrect). This demonstrates that stronger instruction-tuning may yield lower absolute scores beyond the cutoff through more persistent refusal behavior. However, both patterns—consistent low scores and consistent refusals—represent the systematic shifts that PELT reliably detects.

6 Conclusions and Future Work

We introduced LLMLagBench, a benchmark designed to empirically identify temporal knowledge boundaries in LLMs through systematic evaluation of their performance on dated factual events. Our findings demonstrate a number of important insights.

First, knowledge infusion operates differently across training phases resulting in multiple partial cutoff points within a single model rather than a single sharp boundary. Second, smaller models show substantially weaker performance on time-sensitive questions and lower post-cutoff refusal rates, making them more susceptible to hallucination about recent events. Third, provider-declared cutoffs and model self-reports may diverge from empirically detected boundaries, with discrepancies ranging from months to years, underscoring the necessity of independent empirical validation.

In our future work we plan to extend LLMLagBench to evaluate retention of localized knowledge by developing country-specific versions using regional news sources, beginning with Polish news reports. Current version of LLMLagBench will be evaluated on a regular basis.

7 Availability

LLMLagBench is publicly available as an interactive leaderboard at <https://huggingface.co/spaces/pelcra/llmlagbench>. The underlying dataset of time-sensitive questions is withheld to prevent leaks. Model developers and researchers can request evaluations by contacting pelcra@uni.lodz.pl.

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